WEEK 4: ASSIGNMENT 4

Q1 Given:
$\bar{x} = 32$, $\mu_0 = 30$, $\sigma = 10$, $n = 30$, $\alpha = 0.05$
Test: H_0: $\mu \leq 30$ vs H_1: $\mu > 30$
The calculated z-value is closest to:
a) 0.87
b) 1.09
c) 1.64
d) 2.05

Q2 For a one-tailed test at $\alpha = 0.05$, if p-value = 0.137, the correct decision is:
a) Reject H_0
b) Accept H_0
c) Do not reject H_0
d) Insufficient data

Q3 Rejecting a true null hypothesis is called:
a) Type II error
b) Power of the test
c) Type I error
d) Sampling error

Q4 If the sample size is fixed and α is decreased, then β will:
a) Decrease
b) Increase
c) Remain unchanged
d) Become zero

Q5 A sample of 120 accidents shows 67 due to drunk driving.
Test H_0: $p = 0.5$ vs H_1: $p \neq 0.5$
The test statistic z is closest to:
a) 0.98
b) 1.15
c) 1.28
d) 1.64

Q6 If in a z test, the p-value = 0.2006 and $\alpha = 0.05$, the decision is:
a) Reject H_0
b) Not enough evidences
c) Do not reject H_0
d) Increase sample size

Q7 If σ_1^2 and σ_2^2 are known, the appropriate test statistic for $\mu_1 - \mu_2$ is:
a) t statistic
b) χ^2 statistic
c) z statistic
d) F statistic

Q8 Which hypothesis must always contain equality?
a) Alternative hypothesis
b) Research hypothesis
c) Null hypothesis
d) Working hypothesis

Q9 Which action reduces both Type I and Type II errors simultaneously?
a) Increasing α
b) Decreasing α
c) Increasing sample size
d) Changing tail of test

Q10 In the past, 60% of the tourists who visited Delhi went to see Agra City. The management of Agra City recently undertook an extensive promotional campaign. They are interested in determining whether the promotional campaign actually increased the proportion of tourists visiting Agra City. The correct set of hypotheses is
a) H_0: $P > 0.60$ Ha: $P \leq 0.60$
b) H_0: $P < 0.45$ Ha: $P \geq 0.45$
c) H_0: $P \geq 0.45$ Ha: $P < 0.45$
d) H_0: $P \leq 0.60$ Ha: $P > 0.60$

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